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## Postprocessing device that forms air layer between upper face of sheets already stacked on processing tray and lower face of sheet newly placed on processing tray, and image forming system


### Abstract

A postprocessing device is attached to an image forming apparatus. The postprocessing device includes a processing tray, a sheet processor, and a fan. The processing tray is for stacking thereon sheets on which an image has been formed and which is transported from the image forming apparatus. The sheet processor performs predetermined postprocessing on the sheets stacked on the processing tray. The fan forms an air layer between an upper face of the sheets already stacked on the stacking surface of the processing tray and a lower face of a sheet newly placed on the processing tray, by emitting air in a transport direction of the sheet, from an upstream side in the transport direction toward the stacking surface.



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## Background/Summary

INCORPORATION BY REFERENCE
(1) This application claims priority to Japanese Patent Application No. 2022-024999 filed on Feb. 21, 2022, the entire contents of which are incorporated by reference herein.
BACKGROUND
(2) The present disclosure relates to a postprocessing device that performs postprocessing on a plurality of sheets, and an image forming system that includes such postprocessing device.
(3) In an image forming apparatus, an image reading device reads the image of a source document, and an image forming device forms the image of the source document on a recording sheet. The postprocessing device receives the recording sheet having the image of the source document formed thereon from the image forming apparatus, and performs postprocessing on the recording sheet. The postprocessing performed by the postprocessing device includes, for example, a stapling operation including aligning the edges of a plurality of recording sheets and binding the edge of the sheaf of the recording sheets, a punching operation for perforating the end portion of the recording sheet, and an inward folding operation for folding the recording sheet.
(4) The postprocessing device includes a processing tray for stacking and temporarily retaining the recording sheets thereon. The postprocessing device performs predetermined postprocessing, on the recording sheets stacked on the processing tray. For example, a postprocessing device is known that includes a tray for temporarily retaining following recording sheets (second intermediate storage

section), located on the upper side of the processing tray (first intermediate storage section).
SUMMARY
(5) The disclosure proposes further improvement of the foregoing techniques.
(6) In an aspect, the disclosure provides a postprocessing device to be attached to an image forming apparatus. The postprocessing device includes a processing tray, a sheet processor, and a fan. The processing tray is for stacking thereon sheets on which an image has been formed and which is transported from the image forming apparatus. The sheet processor performs predetermined postprocessing on the sheets stacked on the processing tray. The fan forms an air layer between an upper face of the sheets already stacked on the stacking surface of the processing tray and a lower face of a sheet newly placed on the processing tray, by emitting air in a transport direction of the sheet, from an upstream side in the transport direction toward the stacking surface.
(7) In another aspect, the disclosure provides an image forming system including an image forming apparatus and a postprocessing device. The image forming apparatus forms an image on a sheet. The postprocessing device receives the sheet from the image forming apparatus, and performs postprocessing on the sheet. The postprocessing device includes a processing tray, a sheet processor, and a fan. The processing tray is for stacking thereon sheets transported from the image forming apparatus. The sheet processor performs predetermined postprocessing on the sheets stacked on the processing tray. The fan forms an air layer between an upper face of the sheets already stacked on the stacking surface of the processing tray and a lower face of a sheet newly placed on the processing tray, by emitting air in a transport direction of the sheet, from an upstream side in the transport direction toward the stacking surface.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a partially cross-sectional front view of an image forming system including a postprocessing device according to an embodiment of the disclosure;
(2) FIG. **2** is an enlarged and partially cross-sectional front view of the postprocessing device;
(3) FIG. **3**A and FIG. **3**B are schematic cross-sectional views each showing how a recording sheet is placed on a processing tray located in a lower portion of the device;
(4) FIG. **4** is a schematic plan view showing a fan and related parts;
(5) FIG. **5** is a functional block diagram schematically showing an essential internal configuration of an image forming apparatus and the postprocessing device; and
(6) FIG. **6** is a flowchart showing a process of placing the recording sheet.
DETAILED DESCRIPTION
(7) Hereafter, a postprocessing device and an image forming system according to an embodiment of the disclosure will be described, with reference to the drawings. FIG. **1** is a partially cross-sectional front view of the image forming system including the postprocessing device according to the embodiment of the disclosure. The image forming system **100** includes an image forming apparatus **10** that reads an image of a source document and forms the image on a recording sheet (exemplifying the sheet in the disclosure), and the postprocessing device **20** that receives the recording sheet from the image forming apparatus **10**, and performs postprocessing on the recording sheet.
(8) The image forming apparatus **10** is a multifunction peripheral having a plurality of functions, such as copying, printing, scanning, and facsimile transmission. The image forming apparatus **10** includes an image reading device **11** and an image forming device **12**. When a plurality of source documents M are placed on a document tray **1**, the image reading device **11** sequentially draws out the source documents M from the document tray **1** one by one, reads the image of each of the source documents M with an image sensor, and sequentially delivers the source documents M to

the discharge tray **2**, so as to stack on each other. The image reading device **11** converts the analog output from the image sensor to a digital signal, and generates image data representing the image of each of the source documents M.
(9) The image forming device **12** forms the image of the source document M represented by the image data, on the recording sheet P through an ink jet process, each time the image data representing the image of each of the source documents M is inputted. The image forming device **12** includes line heads **15** (exemplifying the ink head in the disclosure) that respectively eject ink of four colors, namely black, cyan, magenta, and yellow. The line heads **15** each eject the ink droplets of the corresponding color onto the recording sheet P, delivered to a conveying unit **4** from a paper feeding device **14** through a first transport route **3**, thereby forming a color image on the recording sheet P.
(10) The conveying unit **4** includes a drive roller **8**, a follower roller **9**, tension roller **5**, and a transport belt **6**. The transport belt **6** is an endless belt stretched around the drive roller **8**, the follower roller **9**, and the tension roller **5**. The drive roller **8** is driven by a motor so as to rotate counterclockwise in FIG. **1**. When the drive roller **8** is made to rotate, the transport belt **6** revolves counterclockwise in FIG. **1**, and the follower roller **9** and the tension roller **5** are each passively made to rotate counterclockwise in FIG. **1**, by the transport belt **6**.
(11) The tension roller **5** serves to maintain the tension of the transport belt **6** at an appropriate level. The transport belt **6** is in contact with an adsorption roller **7**. The adsorption roller **7** electrically charges the transport belt **6**, to thereby electrostatically adsorb the recording sheet P delivered from the paper feeding device **14**, to the transport belt **6**.
(12) A controller **66** of the image forming apparatus **10** (FIG. **5**), to be subsequently described, causes the image forming device **12** to form the image of the source document M on the recording sheet P, and transports the recording sheet P to the postprocessing device **20**, through a relay transport route **18** and a delivery roller **19**. In this case, the controller **66** transports the recording sheet P with the face on which the image of the source document M has been formed oriented upward, in other words in a face-up orientation, to the postprocessing device **20**.
(13) When the recording sheet P is to be delivered to the postprocessing device **20** with the face on which the image of the source document M has been formed oriented downward, in other words in a face-down orientation, the controller **66** performs switchback transport, including transporting the recording sheet P from the relay transport route **18** to the transport roller **16**, once stopping the transport roller **16** and then reversely rotating the same, and returning the recording sheet P to the conveying unit **4** through a second transport route **17**, thereby inverting the front and back faces of the recording sheet P. Then the controller **66** transports such recording sheet P to the postprocessing device **20**, through the relay transport route **18** and the delivery roller **19**.
(14) When the image of the source document M is also to be formed on the back face of the recording sheet P, the controller **66** performs the switchback transport, to return the recording sheet P to the conveying unit **4** through the second transport route **17**, in the inverted orientation. Then the controller **66** causes the image forming device **12** to form the image of the source document M on the back face of the recording sheet P, and transports the recording sheet P to the postprocessing device **20**, through the relay transport route **18** and the delivery roller **19**.
(15) FIG. **2** is a partially cross-sectional front view of the postprocessing device **20**. The postprocessing device **20** includes, in the upper area thereof, two transport roller pairs **21** and **22**, a pair of delivery rollers **23**A and **23**B, an output tray **24**, a processing tray **25**, a stapling device **26**, and a paddle **27**. The processing tray **25** is for stacking and temporarily retaining thereon the recording sheets P. The stapling device **26** serves to bind the end portion of a plurality of recording sheets P stacked on the processing tray **25**.
(16) The postprocessing device **20** includes, in the middle area thereof toward the lower side, three transport roller pairs **31** to **33**, a delivery roller pair **34**, an output tray **35**, a processing tray **36**, a blade **37**, a folding roller pair **38**, a fan **39**, and a sheet sensor **40**. The processing tray **36** is for

stacking and temporarily retaining thereon the recording sheets P. The blade **37** and the folding roller pair **38** inwardly fold the recording sheet P placed on the processing tray **36**. The blade **37** and the folding roller pair **38** exemplify the sheet processor in the disclosure.

(17) The blade **37** is driven by a first driver **71** shown in FIG. **5**, for example constituted of a motor, so as to vertically reciprocate with respect to the recording sheet P placed on the processing tray **36**. In the processing tray **36**, a clearance T is formed for the blade **37** pass through. The processing tray **36** includes a belt moving mechanism **41** for moving the recording sheet P placed on the processing tray **36**. The belt moving mechanism **41** includes a drive roller **42**, a follower roller **43**, a moving belt **44**, and a bent piece **45**.

(18) The moving belt **44** is an endless belt stretched around the drive roller **42** and the follower roller **43**. The drive roller **42** is driven by a motor so as to rotate. When the drive roller **42** is made to rotate, the moving belt **44** revolves. To be more specific, the drive roller **42** is driven by a second driver **72** shown in FIG. **5**, for example constituted of a motor, so as to rotate clockwise or counterclockwise in FIG. **2**.

(19) The moving belt **44** pushes up the recording sheet P placed on the processing tray **36** to the upstream side in the transport direction, by revolving clockwise in FIG. **2**. The moving belt **44** pulls back the recording sheet P placed on the processing tray **36** to the downstream side in the transport direction, by revolving counterclockwise in FIG. **2**. The bent piece **45** is located on the outer circumferential surface of the moving belt **44**. The bent piece **45** is bent toward the upstream side in the transport direction, to prevent the leading edge of the recording sheet P placed on the processing tray **36** from falling off.

(20) At the branch point between the transport roller pair **21** and the transport roller pairs **22** and **31**, a flap to be made to swing, by a driver such as a motor, is provided. A drive controller **76** of the postprocessing device **20**, to be subsequently described, causes the flap to swing, so as to deliver the recording sheet P transported from the image forming apparatus **10**, either to the output tray **24** on the left in FIG. **2**, through a third transport route **51**, or to the output tray **35**, on the lower side in FIG. **2**, through a fourth transport route **52**.

(21) The fan **39** emits air in the transport direction of the recording sheet P, from the upstream side in the transport direction toward the stacking surface of the processing tray **36**, thereby forming an air layer between the upper face of the recording sheet P already placed on the stacking surface of the processing tray **36**, and the lower face of the recording sheet P newly placed on the processing tray **36**. On the processing tray **36**, the recording sheet P is placed at the position that exempts the trailing edge of the recording sheet P placed on the stacking surface from being directly hit by the air emitted from the fan **39**. In other words, the processing tray **36** is located at such a position that, when a maximum number of recording sheets P that can be subjected to the operation of the blade **37** and the folding roller **38** are stacked, the lower face of the air layer formed by the fan **39** is located higher than the upper face of the uppermost sheet of the stacked recording sheets P.

(22) The sheet sensor **40** detects the leading edge of the recording sheet P transported through the fourth transport route **52**. The sheet sensor **40** is, for example, an optical reflective sensor having a light emitting element that emits light toward the recording sheet P, and a photodetector that receives the light reflected by the recording sheet P.

(23) [When Delivering Recording Sheet P to Output Tray **24**]

(24) The drive controller **76** of the postprocessing device **20** transports the recording sheet P delivered from the image forming apparatus **10** through the third transport route **51** using the two transport roller pairs **21** and **22**, and then delivers the recording sheet P to the output tray **24**, through the delivery rollers **23**A and **23**B. Alternatively, the drive controller **76** can receive the recording sheet P delivered from the image forming apparatus **10** on the processing tray **25**, cause the stapling device **26** to bind the end portion of the plurality of recording sheets P stacked on the processing tray **25**, and then deliver the sheaf of the recording sheets P bound as above, from the processing tray **25** to the output tray **24**, through the delivery rollers **23**A and **23**B.

(25) [When Delivering Recording Sheet P to Output Tray **35**]
(26) The drive controller **76** of the postprocessing device **20** transports the recording sheet P delivered from the image forming apparatus **10** through the fourth transport route **52**, using the three transport roller pairs **31** to **33**, and receives the recording sheet P discharged from the fourth transport route **52** on the processing tray **36**. Then the drive controller **76** of the postprocessing device **20** causes the moving belt **44** to revolve clockwise as shown in FIG. **3**A, thereby pushing up the recording sheet P placed on the processing tray **36** to the upstream side in the transport direction, to a predetermined position.
(27) The predetermined position is defined such that the leading edge of the recording sheet P**1** discharged from the fourth transport route **52** can be kept from contacting the trailing edge of the recording sheet P**2** placed on the processing tray **36**, and that the air emitted from the fan **39** does not directly hit the trailing edge of the recording sheet P**2**. The position to which the air is emitted from the fan **39** is defined such that the air does not directly hit the trailing edge of the recording sheet P**2** which has reached the predetermined position, for example on the upper side of the uppermost sheet of the maximum number of recording sheets P that the processing tray **36** can accept, and on the downstream side of the trailing edge of the recording sheet P**2**, in the transport direction.
(28) It is for the purpose of preventing the leading edge of the recording sheet P**1** from being caught by the trailing edge of the recording sheet P**2** and disabled from proceeding further, or preventing the trailing edge of the recording sheet P**2** from being lifted up, that the recording sheet P placed on the processing tray **36** is moved to the position where the trailing edge of the recording sheet P**2** is kept from contacting the leading edge of the recording sheet P**1** discharged from the fourth transport route **52**.
(29) When the recording sheet P**1** is discharged from the fourth transport route **52** and placed on the processing tray **36**, the drive controller **76** causes the moving belt **44**, as shown in FIG. **3**B, to revolve counterclockwise, thereby gradually pulling back the recording sheet P placed on the processing tray **36** toward the downstream side in the transport direction, to the position shown in FIG. **2**.
(30) The drive controller **76** of the postprocessing device **20** moves the sheaf of the recording sheets P placed on the processing tray **36** using the moving belt **44**, so as to bring the position to be folded in the sheaf of the recording sheets P to the clearance T. Then the drive controller **76** moves the blade **37**, so that the blade **37** pushes up the sheaf of the recording sheets P, into a mountain fold shape. The drive controller **76** causes the folding roller pair **38** to catch the mountain fold shape portion of the sheaf of the recording sheets P, and to transport the sheaf of the recording sheets P in the folded state, so that the folded sheaf of the recording sheets P is delivered to the output tray **35**, from the processing tray **36** through the delivery roller **34**.
(31) FIG. **4** is a schematic plan view showing the fan **39** and the related parts. As shown in FIG. **4**, a duct **81** is provided that guides the air emitted from the fan **39** toward the stacking surface of the processing tray **36**. The duct **81** includes an air inlet **82** for sucking the air emitted from the fan **39**, an air outlet **83** for ejecting the air, and an intermediate passage **84** connecting between the air inlet **82** and the air outlet **83**. The intermediate passage **84** includes a main passage portion **85** connected to the air inlet **82**, and a sub passage portion **86** branched from the main passage portion **85** and connecting between the main passage portion **85** and the air outlet **83**. The sub passage portion **86** is smaller in cross-sectional area for emitting the air, than the main passage portion **85**. The portion in the intermediate passage **84** communicating with the air outlet **83** (i.e., sub passage portion **86**) extends parallel to the transport direction of the recording sheet P (longitudinal direction orthogonal to the width direction).
(32) Hereunder, a configuration related to the control operation of the image forming apparatus **10** and the postprocessing device **20** will be described. FIG. **5** is a functional block diagram schematically showing an essential internal configuration of the image forming apparatus **10** and

the postprocessing device **20**, constituting the image forming system **100**. As shown in FIG. **5**, the image forming apparatus **10** includes the image reading device **11**, the image forming device **12**, a display device **61**, an operation device **62**, a touch panel **63**, a storage device **64**, the controller **66**, and an interface (I/F) **67**. The mentioned components are configured to transmit and receive data and signals to and from each other, via a bus.
(33) The display device **61** is, for example, constituted of a liquid crystal display (LCD) or an organic light-emitting diode (OLED) display. The operation device **62** includes physical keys such as a tenkey, an enter key, and a start key. The operation device **62** receives inputs of various instructions, corresponding to the user's operation performed on the mentioned keys.
(34) A touch panel **63** is overlaid on the screen of the display device **61**. The touch panel **63** is based on a resistive film or electrostatic capacitance. The touch panel **63** detects a contact (touch) of the user's finger made thereon, along with the touched position, and outputs a detection signal indicating the coordinate of the touched position, to the control device **66**.
(35) The storage device **64** is a large-capacity storage device such as a solid-state drive (SSD) or a hard disk drive (HDD). The storage device **64** contains various application programs and various types of data.
(36) The controller **66** includes a processor, a random-access memory (RAM), a read-only memory (ROM), and so forth. The processor is, for example, a central processing unit (CPU), an application specific integrated circuit (ASIC), or a micro processing unit (MPU). The controller **66** serves as a processing device that executes the control program stored in the ROM or the storage device **64**, thereby executing various processings necessary for the image forming job by the image forming apparatus **10**.
(37) The controller **66** is connected to the image reading device **11**, the image forming device **12**, the display device **61**, the operation device **62**, the touch panel **63**, the storage device **64**, and the I/F **67**. The controller **66** controls the operation of the components cited above, and transmits and receives signals and data to and from those components.
(38) For example, the controller **66** controls the displaying operation of the display device **61**. The controller **66** receives the instruction inputted by the user, on the basis of the detection signal outputted from the touch panel **63** or a press of the physical key on the operation device **62**. For example, the controller **66** receives the instruction according to a touch operation, performed through the touch panel **63** on the graphical user interface (GUI) displayed on the screen of the display device **61**.
(39) The postprocessing device **20** includes the fan **39**, the sheet sensor **40**, the first driver **71** to a fourth driver **74**, the drive controller **76**, and an I/F **77**. These components are configured to transmit and receive data and signals to and from each other, via a bus.
(40) The drive controller **76** includes a processor, a RAM, a ROM, and so forth. The processor is, for example, a central processing unit (CPU), an application specific integrated circuit (ASIC), or a micro processing unit (MPU). The drive controller **76** serves as a processing device that executes the drive control program stored in the ROM, thereby executing various operations necessary for the postprocessing by the postprocessing device **20**.
(41) The controller **66** of the image forming apparatus **10** and the drive controller **76** of the postprocessing device **20** are configured to input and output data and signals between each other, via the respective I/Fs **67** and **77**. For example, the controller **66** of the image forming apparatus **10** outputs a control signal for instructing the postprocessing device **20** to perform the postprocessing, to the drive controller **76** of the postprocessing device **20**. The drive controller **76** of the postprocessing device **20** controls the fan **39**, the first driver **71**, and the second driver **72**, according to the control signal received.
(42) The first driver **71** and the second driver **72** of the postprocessing device **20** each include, for example, a stepping motor serving as the drive source. As already described, for example the first driver **71** includes the drive source for moving the blade **37**. Likewise, the second driver **72**

includes the drive source for rotating the drive roller **42**.
(43) The drive controller **76** controls the action of the motor of the first driver **71**, thereby causing the blade **37** to vertically reciprocate, with respect to the recording sheet P placed on the processing tray **36**. The drive controller **76** controls the action of the motor of the second driver **72**, so as to cause the drive roller **42** to rotate clockwise or counterclockwise, thereby causing the moving belt **44** to revolve. With such operation, the moving belt **44** can push up the recording sheet P placed on the processing tray **36** to the upstream side in the transport direction, and move the recording sheet P**2** to the downstream side in the transport direction.
(44) Referring now to a flowchart shown in FIG. **6**, a control operation performed by the drive controller **76** of the postprocessing device **20**, to place the recording sheet P on the processing tray **36** in the lower area of the device, will be described hereunder.
(45) Upon receipt of an instruction to execute the inward folding operation, according to the user's operation performed on the start key of the operation device **62**, the controller **66** of the image forming apparatus **10** outputs a control signal indicating the instruction to execute the inward folding operation, to the postprocessing device **20** through the I/F **67**.
(46) Upon receipt of the control signal indicating the instruction to execute the inward folding operation, through the I/F **77**, the drive controller **76** of the postprocessing device **20** controls the second driver **72** so as to cause the drive roller **42** to rotate clockwise in FIG. **3**A thus to cause the moving belt **44** to revolve clockwise, thereby pushing up the recording sheet P placed on the processing tray **36** to the upstream side in the transport direction, as shown in FIG. **3**A (step S**1**). Here, since no recording sheet is placed on the processing tray **36**, when the first recording sheet P is transported, the operation of step S**1** may be started when the second recording sheet P is about to be transported.
(47) The drive controller **76** controls the fan **39**, so as to emit the air from the fan **39** to the stacking surface of the processing tray **36** (step S**2**). The drive controller **76** controls the respective drive sources of the transport roller pairs **31** to **33**, so as to transport the recording sheet P delivered from the image forming apparatus **10**, through the fourth transport route **52** (step S**3**). Upon deciding that the recording sheet P**1** has been discharged from the fourth transport route **52**, and the leading edge of the recording sheet P**1** has passed the position under the sheet sensor **40**, according to the detection output from the sheet sensor **40** (YES at step S**4**), the drive controller **76** controls the second driver **72**, so as to cause the drive roller **42** to rotate counterclockwise in FIG. **3**B thus to cause the moving belt **44** to revolve counterclockwise, thereby pulling back the recording sheet P placed on the processing tray **36** to the downstream side in the transport direction, as shown in FIG. **3**B (step S**5**).
(48) The drive controller **76** repeats the operation of step S**1** to step S**5**, until a prespecified number of recording sheets P are all stacked on the processing tray **36**. When the prespecified number of recording sheets P are all stacked on the processing tray **36**, the drive controller **76** controls the moving belt **44**, the blade **37**, and the folding roller **38** by known control methods, so as to inwardly fold the sheaf of the recording sheet P, and deliver the folded sheaf of the recording sheets P to the output tray **35**.
(49) Now, since a plurality of recording sheets P are stacked on the processing tray **36**, the upper face of the recording sheet P**2** already placed on the processing tray **36**, and the lower face of the recording sheet P**1** being newly placed enter into contact with each other, during the transport operation of the recording sheets P. Such a contact may impede the recording sheet P from proceeding further, or cause the recording sheet P to bounce, because of the friction between the recording sheets P, which may lead to failure in properly placing the recording sheets P on the processing tray **36**. In particular, the recording sheets P subjected to the ink jet printing operation are often transported in a wet state, with the ink undried yet, and therefore the frictional force between the recording sheets P is increased.
(50) Although the aforementioned known postprocessing device can reduce the friction arising

from the contact, it takes a longer time before the recording sheet P is placed on the processing tray, and therefore the productivity is lowered.
(51) With the configuration according to the foregoing embodiment, in contrast, the air emitted from the fan **39** forms the air layer, between the upper face of the recording sheet P**2** already placed on the stacking surface of the processing tray **36**, and the lower face of the recording sheet P**1** newly placed on the processing tray **36**.
(52) Without such air layer, for example, the leading edge of the recording sheet P**1** delivered through the transport roller **33** to be newly placed on the processing tray **36** makes contact with the upper face of the recording sheet P**2** already placed on the processing tray **36**. In this case, the recording sheet P**1** is bent in the region between the leading edge thereof and the exit **521** of the transport route **331**, and the upper face of the recording sheet P**2** already placed on the processing tray **36** is biased to the downstream side in the transport direction, by the leading edge of the recording sheet P**1**, owing to the restoring force (stiffness) of the recording sheet P**1** to recover from the bent state. As result, the stacking status of the sheaf of the recording sheets P**2** may become irregular, or the recording sheet P**1** may fail to be properly placed on the sheaf of the recording sheets P**2**.
(53) However, the air layer formed by the air emitted from the fan **39**, as in the foregoing embodiment, serves to mitigate the extent of the bending of the recording sheet P**1**, delivered through the transport roller **33** to be newly placed on the processing tray **36**, when the leading edge thereof is about to contact the upper face of the recording sheet P**2** already placed on the processing tray **36**. Accordingly, the biasing force exerted by the leading edge of the recording sheet P**1**, attempting to move the upper face of the recording sheet P**2** to the downstream side in the transport direction, can be minimized. Therefore, the recording sheet P**1** can be placed on the recording sheet P**2**, with reduced chance that the upper face of the recording sheet P**2** and the lower face of the recording sheet P**1** make contact with each other. As result, the friction arising from the contact can be reduced, and the recording sheets P stacked on the processing tray **36** can be properly aligned, which leads to minimized degradation in productivity.
(54) According to the foregoing embodiment, the portion of the duct **81** communicating with the air outlet **83** (sub passage portion **86**) extends parallel to the transport direction of the recording sheet P (longitudinal direction orthogonal to the width direction), as shown in FIG. **4**. Such a configuration enables the air emitted from the fan **39** to hit the recording sheet P placed on the processing tray **36**, in the direction parallel to the longitudinal direction of the recording sheet P. Therefore, the recording sheet P can be prevented from being displaced in the transport direction or in the width direction.
(55) According to the foregoing embodiment, further, the air outlet **83** of the duct **81** is located at the position where the trailing edge of the recording sheet P**2** is exempted from being directly hit by the air from the fan **39**, when the recording sheet P**2** already placed on the processing tray **36** is moved to the upstream side in the transport direction, and the next recording sheet P**1** is about to be placed on the processing tray **36**. Such a position corresponds, for example, to the position downstream of the trailing edge of the recording sheet P**2** in the transport direction, and higher than the upper surface of the uppermost sheet of the maximum number of recording sheets P that the processing tray **36** can accept. Such a configuration prevents the trailing edge of the recording sheet P from being lifted up by the air from the fan **39**.
(56) When the recording sheet P loses rigidity (becomes less stiff) owing to moisture, the posture of the recording sheet P becomes unstable, which may affect the transport performance. However, the recording sheet P can be subjected to the wind pressure generated by the air from the fan **39** in the foregoing embodiment, thus to be facilitated to maintain the correct posture. Therefore, when delivering the recording sheet P to the processing tray **36**, it is preferable to activate the fan **39** to emit the air, even though no recording sheet P is placed on the processing tray **36** yet.
(57) As another embodiment, the drive controller **76** may change the drive control of the fan **39**, on

the basis of predetermined information related to the printing operation, or the instruction from the user related to the printing operation. The predetermined information may include, for example, print coverage, the type of the sheet, and the printing method.
(58) When the print coverage is low, or when the sheet is thick, the rigidity of the recording sheet P is barely lowered. Accordingly, upon deciding that the print coverage is equal to or lower than a predetermined first threshold, or that the thickness of the sheet is equal to or thicker than a predetermined second threshold, on the basis of the predetermined information transmitted from the image forming apparatus **10**, the drive controller **76** may keep from activating the fan **39**, or reduce the number of rotations of the fan **39** per unit time (that is, rotational speed) by a predetermined value, thereby reducing the air volume from the fan **39**.
(59) When the image is formed by electrophotography, using a developing agent such as a toner, the recording sheet P becomes less wet, compared with the case of the ink jet printing, and therefore the rigidity of the recording sheet P is barely lowered. Accordingly, the drive controller **76** may keep from activating the fan **39**, for example when a signal indicating that the printing method is the electrophotography is received from the image forming apparatus **10**, as the predetermined information, but may activate the fan **39**, when a signal indicating that the printing method is the ink jet printing is received from the image forming apparatus **10**, as the predetermined information.
(60) Further, for example when the user inputs a first instruction for cancelling the operation of the fan **39**, through the operation device **62**, the image forming apparatus **10** transmits a signal representing the first instruction, to the postprocessing device **20**. When the postprocessing device **20** receives the signal representing the first instruction, the drive controller **76** may keep from activating the fan **39**. In addition, when the user inputs a second instruction for degrading the performance of the fan **39**, through the operation device **62**, the image forming apparatus **10** transmits a signal representing the second instruction, to the postprocessing device **20**. When the postprocessing device **20** receives the signal representing the second instruction, the drive controller **76** may reduce the number of rotations of the fan **39** per unit time by a predetermined value, thereby reducing the air volume from the fan **39**.
(61) The disclosure is not limited to the configuration according to the foregoing embodiment, but may be modified in various manners. For example, although the disclosure is applied to the processing tray **36** for performing the inward folding operation in the embodiment, the disclosure may be applied to a processing tray for performing a different postprocessing, such as the stapling operation.
(62) Further, the configurations and processings described in the foregoing embodiment and variations with reference to FIG. **1** to FIG. **6** are merely exemplary, and in no way intended to limit the disclosure to those configurations and processings.
(63) While the present disclosure has been described in detail with reference to the embodiments thereof, it would be apparent to those skilled in the art the various changes and modifications may be made therein within the scope defined by the appended claims.

## Claims

1. A postprocessing device to be attached to an image forming apparatus, the postprocessing device comprising: a processing tray for stacking thereon sheets on which an image has been formed, and which is transported from the image forming apparatus; a sheet processor that performs predetermined postprocessing on the sheets stacked on the processing tray; a fan that forms an air layer between an upper face of the sheets already stacked on a stacking surface of the processing tray and a lower face of a sheet newly placed on the processing tray, by emitting air in a transport direction of the sheet, from an upstream side in the transport direction toward the stacking surface; and a duct that guides the air emitted from the fan toward the stacking surface, wherein the duct includes an air inlet for sucking the air emitted from the fan, a plurality of air outlets for ejecting

the air, and an intermediate passage connecting between the air inlet and the plurality of air outlets, and the intermediate passage includes a main passage portion connected to the air inlet, and a plurality of sub passage portions connecting between the main passage portion and each of the plurality of air outlets, and each of the plurality of sub passage portions is smaller in cross-sectional area for emitting the air than the main passage portion, and extends parallel to the transport direction of the sheet, wherein the postprocessing device further comprising a drive controller configured to control an action of the fan, by executing a control program, and the drive controller is configured to change drive control of the fan, on a basis of predetermined information related to a printing operation by the image forming apparatus, or an instruction from a user related to the printing operation, wherein the postprocessing device further comprising an interface device that communicates with the image forming apparatus, the drive controller keeps from activating the fan, when a signal indicating that the printing method is the electrophotography, as the predetermined information, is received from the image forming apparatus via the interface device, and the drive controller activates the fan, when a signal indicating that the printing method is the ink jet printing, as the predetermined information, is received from the image forming apparatus via the interface device.

2. The postprocessing device according to claim 1, wherein the processing tray is located at a position that exempts a trailing edge of the sheets stacked on the stacking surface from being directly hit by the air emitted from the fan.

3. The postprocessing device according to claim 1, wherein the plurality of air outlets are located at a position higher than an upper surface of an uppermost sheet of the sheets, in a case where a maximum number of sheets that the sheet processor can accept are stacked on the processing tray.

4. A postprocessing device to be attached to an image forming apparatus, the postprocessing device comprising: a processing tray for stacking thereon sheets on which an image has been formed, and which is transported from the image forming apparatus; a sheet processor that performs predetermined postprocessing on the sheets stacked on the processing tray; a fan that forms an air layer between an upper face of the sheets already stacked on a stacking surface of the processing tray and a lower face of a sheet newly placed on the processing tray, by emitting air in a transport direction of the sheet, from an upstream side in the transport direction toward the stacking surface; and a drive controller configured to control an action of the fan, by executing a control program, wherein the drive controller is configured to change drive control of the fan, on a basis of predetermined information related to a printing operation by the image forming apparatus, or an instruction from a user related to the printing operation, wherein the predetermined information includes a print coverage and a thickness of the sheet, and the drive controller keeps from activating the fan, or reduces a rotational speed by a predetermined value, when the print coverage is equal to or lower than a predetermined first threshold, or when the thickness of the sheet is equal to or thicker than a predetermined second threshold.

5. An image forming system comprising: an image forming apparatus that forms an image on a sheet; and a postprocessing device according to claim 4.